LinkedIn 数据

方法说明

介绍

本方法论概述定义了为了解和监控人工智能劳动力趋势而开发的几个 LinkedIn 指标。这组作品代表了通过 LinkedIn 数据看到的世界，这些数据来自 LinkedIn 全球 1+ 亿会员的匿名和汇总个人资料信息。因此，它受到会员选择如何使用该平台的影响，该平台可能会根据专业、社会和区域文化以及整体网站可用性和可访问性而有所不同。

国家样本

纳入的国家/地区是符合条件的国家/地区的精选样本，这些国家/地区 LinkedIn 的劳动力覆盖率至少为 40%，并且在任何给定月份至少雇用了 10 名 AI。印度被纳入该样本是因为它在全球经济中的重要性日益增加，但其 LinkedIn 覆盖率并未达到 40% 的劳动力。对印度的见解可能无法像其他国家那样提供全面的情况，应该进行相应的解释。

人工智能技能

LinkedIn 会员在他们的 LinkedIn 个人资料上自我报告他们的技能。目前，LinkedIn 识别了超过 41,000 种不同的标准化技能。

LinkedIn 将 AI 技能分为 2 个相互排斥的组：“AI 工程”和“AI 素养”技能，其中广义上，AI 工程技能是指设计、开发、部署和维护人工智能系统所需的技术专长和实践能力，而 AI 素养技能是指理解、评估和有效与 AI 技术交互所需的知识、能力和批判性思维能力。随着技能的不断发展，LinkedIn 会定期维护和更新这些分类。

技能分组是由专业分类学家通过相似性指数方法得出的，该方法衡量行业层面的技能构成。根据 ISIC 4 行业分类对行业进行分类（Zhu et al.， 2018）。

构成“人工智能工程”技能组的顶级技能是机器学习、自然语言处理、人工智能、计算机视觉、图像处理、深度学习、TensorFlow 和 OpenCV 等。构成“人工智能素养”组的技能包括提示工程、GPT-3、GPT-4、稳定扩散、ChatGPT、Github 副驾驶、生成艺术、Dall-E、Midjourney 和 Google Bard。

人工智能职业

LinkedIn 会员头衔是标准化的，分为 16,000 多个职业。这些不是针对特定部门或国家/地区的。人工智能工作是一种需要人工智能技能才能完成工作的职业。此类职业的例子包括（但不限于）：机器学习工程师、人工智能专家、数据科学家、计算机视觉工程师等。

AI人才

如果 LinkedIn 会员在其个人资料中明确添加了至少两项 AI 技能和/或他们正在或曾经受雇于 AI 工作，则该会员被视为 AI 人才。

方法论

技能基因组

对于任何实体（职业或工作、国家、部门等），技能基因组是该实体的 50 项“最具特征的技能”的有序列表（载体）。TF-IDF算法（“术语频率-逆文档频率”的缩写）用于识别每个实体最具代表性的技能，并降低对有关该特定实体（例如Microsoft Word）的信息很少的无处不在的技能进行排名。

TF-IDF 是一种统计指标，用于评估一个单词对文档集合或语料库中的文档的重要性（Rajaraman 和 Ullman，2011 年）。在这种情况下，TF-IDF 算法用于评估技能对所选实体的代表性。这是通过将两个指标相乘来完成的：

1. 实体中技能的术语频率（“TF”）。
2. 技能在一组实体中的对数反实体频率（“IDF”）。这表示单词在整个实体集中的常见程度或罕见程度。IDF 越接近 0，单词越常见。

如果该技能在 LinkedIn 实体中非常普遍，并且出现在许多职位或成员描述中，则 IDF 将接近 0。另一方面，如果该技能是特定实体所独有的，以色列国防军将接近 1。更多详细信息，请访问 LinkedIn 的技能基因组和 LinkedIn-世界银行方法论。

AI技能渗透

该指标的目的是通过以下方法衡量一个实体（在特定国家、行业、性别等）中人工智能技能的强度：

- 计算 2015-2023 年给定实体（职业、行业等）中 LinkedIn 成员的所有自添加技能的频率。
- 使用 TF-IDF 模型重新加权技能频率，以获得该实体中最具代表性的前 50 个技能。这 50 项技能构成了该实体的“技能基因组”。
- 计算所选实体中排名靠前的技能中属于 AI 技能组的技能份额。

解释： 人工智能技能渗透率表明人工智能技能在各个职业中的普遍性，或者 LinkedIn 会员在工作中利用人工智能技能的强度。例如，“工程师”职业的前 50 项技能是根据它们在 LinkedIn 成员个人资料中出现的加权频率计算得出的。如果工程师拥有的技能中有四项属于人工智能技能组，那么这一衡量标准表明人工智能技能在工程师中的渗透率估计为8%（即4/50）。

Relative AI skills penetration

To allow for skills penetration comparisons across countries, the skills genomes are calculated and compared to the global average. A ratio is then constructed between a country’s AI skills penetrations and the global average, controlling for occupations.

Note that a country’s AI penetration is benchmarked by considering only the overlapping occupations between this country and the global average set. For example, if Peru has only 2 occupations with AI skills, we calculate the global average using only those two occupations.

Interpretation: A country's relative AI skills penetration of 1.5 indicates that AI skills are 1.5 times as frequent as the global average, for an overlapping set of occupations.

Relative AI skills penetration by country

For cross-country comparison, we present the relative penetration rate of AI skills, measured as the sum of the penetration of each AI skill across occupations in a given country, divided by the average global penetration of AI skills across the overlapping occupations in a sample of countries.

Interpretation: A relative penetration rate of 2 means that the average penetration of AI skills in that country is two times the global average across the same set of occupations.

Relative AI skills penetration by industry

The relative AI skills penetration by country and industry provides an in-depth sectoral decomposition of AI skill penetration across industries and sample countries.

Interpretation: A country's relative AI skill penetration rate of 2 in the education sector means that the average penetration of AI skills in that country is two times the global average across the same set of occupations in that sector.

Relative AI skills penetration by gender

The relative AI skills penetration by country and gender provides an in-depth sectoral decomposition of AI skill penetration across genders and sample countries.

Interpretation: A country's relative AI skill penetration rate for women equal to 2 means that the average penetration of AI skills among women in that country is 2 times the global average for women across the same set of occupations.

Relative AI skills penetration: country rankings over time

The ranking is calculated by estimating the ratio between a country's AI skills penetration and the average AI skills penetration of all countries in the sample, controlling for occupations.

Top AI skills worldwide

AI skills most frequently added by members from 2015 onwards.

Fastest growing AI skills

Top 10 fastest growing "AI engineering" and "AI literacy" skills (year-over-year growth rates) in LinkedIn member profiles globally.

AI jobs or occupations

LinkedIn member titles are standardised and grouped into approximately 15 000 occupations. These are not sector or country specific. These occupations are further standardised into approximately 3 600 occupation representatives or 'jobs'. Occupation representatives group occupations with a common role and specialty, regardless of seniority.

An **'AI job'** is an occupation representative that requires AI skills to perform the job. Skills penetration is used as a signal for whether AI skills are prevalent in an occupation representative, in any sector where the occupation representative may exist. Examples of such occupations include Machine Learning Engineer, Artificial Intelligence Specialist, Data Scientist, and Computer Vision Engineer.

AI talent

A LinkedIn member is considered an AI talent if they are occupied in an AI job.

AI talent concentration

The counts of AI talent are used to calculate talent concentration metrics. For example, AI talent concentration at the country level is calculated using the counts of AI talent vis-a-vis the counts of LinkedIn members in that country. As such, AI talent concentration metrics may be influenced by a country's LinkedIn coverage and should be used with caution.

Since it also encompasses LinkedIn members with AI job titles – as opposed to only LinkedIn members with AI skills on their profiles – AI talent is considered to be a more comprehensive measure than AI skills.

The aggregates displayed in the "AI talent concentration by industry and gender" chart include data from G20 countries, OECD member countries, Cyprus, Hong Kong (China), Romania, Singapore, United Arab Emirates, and Uruguay.

AI talent migration

Data on AI skills migration comes from the World Bank Group-LinkedIn "Digital Data for Development" partnership. Please see https://linkedindata.worldbank.org/ and Zhu et al. (2018) for more information.

LinkedIn migration rates are derived from the self-identified locations of LinkedIn member profiles. For example, when a LinkedIn member updates his or her location from Paris to London, this is counted as a migration. Migration data is available from 2019 onwards.

LinkedIn data provide insights to countries on the AI talent gained or lost due to migration trends. AI Talent migration is considered for all members with AI skills/holding AI jobs at time t for country A as the country of interest and country B as the source of inflows and destination for outflows. Thus, net AI Talent migration between country A and country B – for country A – is calculated as follows:

$$Net\ AI\ Talent\ Migration_{a,b,t} = \frac{Net\ AI\ Talent\ flows_{a,b,t}}{Member\ count_{a,t}}$$

Net flows are defined as total arrivals minus departures within the given time period. LinkedIn membership varies considerably between countries, which makes interpreting absolute movements of members from one country to another difficult. To compare migration flows between countries fairly, migration flows are normalised for the country of interest. For example, if country A is the country of interest, all absolute net flows into and out of country A, regardless of origin and destination countries, are normalised based on LinkedIn membership in country A at the end of each year and multiplied by 10 000. Hence, this metric indicates relative talent migration from all countries to and from country A.

From 2023 onwards, AI skills have been divided into “AI engineering” and “AI literacy” skills, as described above. Only the AI engineering skills are considered to compute AI talent migration indicators.

Relative AI hiring index

The AI hiring over time chart on OECD.AI indicates the rate of hiring in the AI field for each country, compared to the overall hiring in that country.

The LinkedIn hiring rate is a measure of hires normalised by LinkedIn membership. It is computed as the percentage of LinkedIn members who added a new employer in the same period the job began, divided by the total number of LinkedIn members in the corresponding location.

The AI hiring rate is computed utilising the overall hiring rate methodology, but only considering members classified as AI talent. The relative AI talent hiring rate YoY ratio is the year-over-year change in the AI hiring rate relative to the overall hiring rate in the same country. We share a 12-month moving average.

Interpretation: The Relative AI Hiring Index shows how fast each country is experiencing growth in AI talent hiring relative to growth in overall hiring in the country. A ratio of 1.2 means the growth in AI talent hiring has outpaced the growth in overall hiring by 20%.

Career transitions into AI jobs

The data on career transition shows the share of transitions from any job occupation into AI jobs pooled over a five-year period. Career transitions are computed by aggregating member-level job transitions from one occupation to another occupation that the member has never previously held. To note, first occupations by new graduates and transitions within the same occupation are excluded from this indicator.

LinkedIn member titles are standardised and grouped into over 16,000 occupations. For the purpose of simplifying the career transition visualisations, some of these occupations have been grouped into broader categories:

- Machine learning engineer, machine learning specialist, machine learning researcher and machine learning consultant as machine learning expert;
- Data science specialist and data science researcher as data scientist;

- Deep learning engineer and deep learning researcher as deep learning specialist;
- Data miner and data mining engineer as data mining specialist;
- Artificial intelligence engineer and artificial intelligence researcher as artificial intelligence specialist.

Interpretation: In the United-States, 26.9% of transitions into AI Engineer came from Software Engineers. *Please note that minimum thresholds have been applied, this indicator only captures countries which have at least 3 country/occupation transition pairs and includes transitions with sufficient sample size.*

Proportion of AI talent by gender

The proportion of AI talent by gender shows the percentage of AI talents which are female and male individuals. This indicator is computed yearly at a global and country level and includes industry aggregates.

Interpretation: The share of female AI talent with AI Engineering skills is 30.5% globally in 2024, up from 29.3% in 2016. *Please note that this indicator only includes countries with more than 100 thousand LinkedIn members and that have a gender coverage which is at least of 67%.*

For more information on the methodological note please see LinkedIn Economic Graph's technical note on the methodology used for LinkedIn's AI Data Partnerships.

References

Rajaraman, A. and Ullman, J. (2011), Mining of Massive Datasets, pp. 1–17, https://doi.org/10.1017%2FCBO9781139058452.002, ISBN 978-1-139-05845-2.

Zhu, T.; Fritzler, A.; and Orlowski, J. (2018). World Bank Group-LinkedIn Data Insights: Jobs, Skills and Migration Trends Methodology and Validation Results (English). Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/827991542143093021/World-Bank-Group-LinkedIn-Data-Insights-Jobs-Skills-and-Migration-Trends-Methodology-and-Validation-Results.